

Lecture 4: Parameter estimation

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1 The Framework of Estimation

1.1 Introduction to Statistical Inference

In probability theory, we assume the parameters of a distribution (e.g., p of a Bernoulli, μ, σ^2 of a Normal) are known. In statistics, we face the inverse problem: we have a set of observed data, and we wish to infer the parameters of the distribution that generated it.

- **Parameter:** A numerical characteristic of a population distribution (e.g., μ, p, λ). Parameters are typically unknown and considered fixed constants in the frequentist framework.
- **Statistic:** Any function of the data from a sample. A statistic is a random variable because its value depends on the particular random sample we happen to draw.
- **Estimator:** A statistic whose purpose is to estimate an unknown parameter. It is a function or rule. Example: $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i$.
- **Estimate:** A specific value of an estimator, computed from a particular sample. Example: $\hat{\mu} = 175.3$ cm.

1.2 Method of Moments (MoM)

The Method of Moments is an intuitive first approach to finding estimators. The principle is to equate the sample moments to the theoretical moments of the distribution and solve for the unknown parameters.

The k -th **theoretical** moment is $m_k = \mathbb{E}[X^k]$. The k -th **sample** moment is $\hat{m}_k = \frac{1}{n} \sum_{i=1}^n X_i^k$. We set $m_k = \hat{m}_k$ and solve the equation.

Bernoulli Distribution: Let $X_1, \dots, X_n$ be i.i.d. Bernoulli(p). We wish to estimate p .

- **Theoretical Moment:** The first theoretical moment of $X \sim \text{Bernoulli}(p)$ is $m_1 = \mathbb{E}[X] = p$.
- **Sample Moment:** The first sample moment is $\hat{m}_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$.
- **Equating:** We set $m_1 = \hat{m}_1$, which gives the estimator:

$$\hat{p}_{\text{MoM}} = \bar{X}_n.$$

This is very intuitive: our estimate for the probability of success is the proportion of successes in our sample.

Exponential Distribution: Let $X_1, \dots, X_n$ be i.i.d. $\text{Exp}(\lambda)$. We wish to estimate the rate λ .

- **Theoretical Moment:** The first theoretical moment of $X \sim \text{Exp}(\lambda)$ is $\mathbb{E}[X] = 1/\lambda$.
- **Sample Moment:** The first sample moment is $\bar{X}_n$.
- **Equating:** We set $1/\lambda = \bar{X}_n$. Solving for λ gives the estimator:

$$\frac{1}{\hat{\lambda}_{\text{MoM}}} = \bar{X}_n \implies \hat{\lambda}_{\text{MoM}} = \frac{1}{\bar{X}_n}.$$

The estimated rate is the reciprocal of the sample mean waiting time.

1.3 The Principle of Maximum Likelihood

The Method of Moments is intuitive, but the most popular method for parameter estimation is Maximum Likelihood Estimation. The core idea is: **What parameter value θ makes the data we observed the most probable?**

Definition 1 (Likelihood and Log-Likelihood). Let $X_1, \dots, X_n$ be an independent sample from a distribution with PDF or PMF $f(x; \theta)$. The **Likelihood Function** is the joint density of the observed data, viewed as a function of the parameter θ :

$$L(\theta|x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta).$$

The **Maximum Likelihood Estimator (MLE)** $\hat{\theta}_{MLE}$ is the value of θ that maximizes $L(\theta)$. In practice, it is easier to maximize the **Log-Likelihood Function**, $\ell(\theta) := \log L(\theta)$, since the logarithm is a monotonic function.

$$\ell(\theta) = \sum_{i=1}^n \log f(x_i; \theta).$$

Bernoulli Distribution: Suppose we observe n trials with $\sum_{i=1}^n x_i = k$ successes. The PMF is $f(x; p) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$. The likelihood function is:

$$L(p|\mathbf{x}) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}(1-p)^{n-\sum x_i} = p^k(1-p)^{n-k}.$$

The log-likelihood is:

$$\ell(p) = \log(p^k(1-p)^{n-k}) = k \log(p) + (n-k) \log(1-p).$$

To maximize, we differentiate and set to zero:

$$\frac{d\ell}{dp} = \frac{k}{p} - \frac{n-k}{1-p} = 0 \implies k(1-p) = p(n-k) \implies k - kp = np - kp \implies \hat{p}_{MLE} = \frac{k}{n} = \bar{X}_n.$$

Normal Distribution: Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$. The log-likelihood is:

$$\ell(\mu, \sigma^2) = \sum_{i=1}^n \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x_i - \mu)^2}{2\sigma^2} \right) \right) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Solving $\frac{\partial \ell}{\partial \mu} = 0$ and $\frac{\partial \ell}{\partial \sigma^2} = 0$ gives the MLEs:

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n \quad \text{and} \quad \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Proposition 1 (Invariance Property of MLEs). If $\hat{\theta}$ is the MLE of θ , then for any function g , the MLE of $g(\theta)$ is $g(\hat{\theta})$. For example, the MLE of the standard deviation σ is $\hat{\sigma}_{MLE} = \sqrt{\hat{\sigma}_{MLE}^2}$.

Proof Sketch (for invertible g). Let $\hat{\theta}$ be the value that maximizes the likelihood $L(\theta)$. Let $\eta = g(\theta)$, so that $\theta = g^{-1}(\eta)$. We can define a new likelihood function for η , called the profile likelihood, $L^*(\eta) = L(g^{-1}(\eta))$. To find the MLE for η , we must find the value $\hat{\eta}$ that maximizes $L^*(\eta)$. Since $L^*(\eta)$ is just a reparameterization of $L(\theta)$, its maximum value must be the same as the maximum of $L(\theta)$. This occurs when its argument $g^{-1}(\eta)$ is equal to $\hat{\theta}$. So, we must have $g^{-1}(\hat{\eta}) = \hat{\theta}$, which implies $\hat{\eta} = g(\hat{\theta})$. $\square$

2 Evaluating Estimators

How do we know if an estimator is “good”? We evaluate it based on several key criteria.

2.1 Bias

Definition 2 (Bias). The bias of an estimator $\hat{\theta}$ is the difference between its expected value and the true parameter value: $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. An estimator is **unbiased** if its bias is zero, i.e., $\mathbb{E}[\hat{\theta}] = \theta$.

Sample Variance: The MLE for the variance of a Normal distribution, $\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$, is **biased**. The commonly used **unbiased sample variance** includes Bessel’s correction:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \implies \mathbb{E}[S^2] = \sigma^2.$$

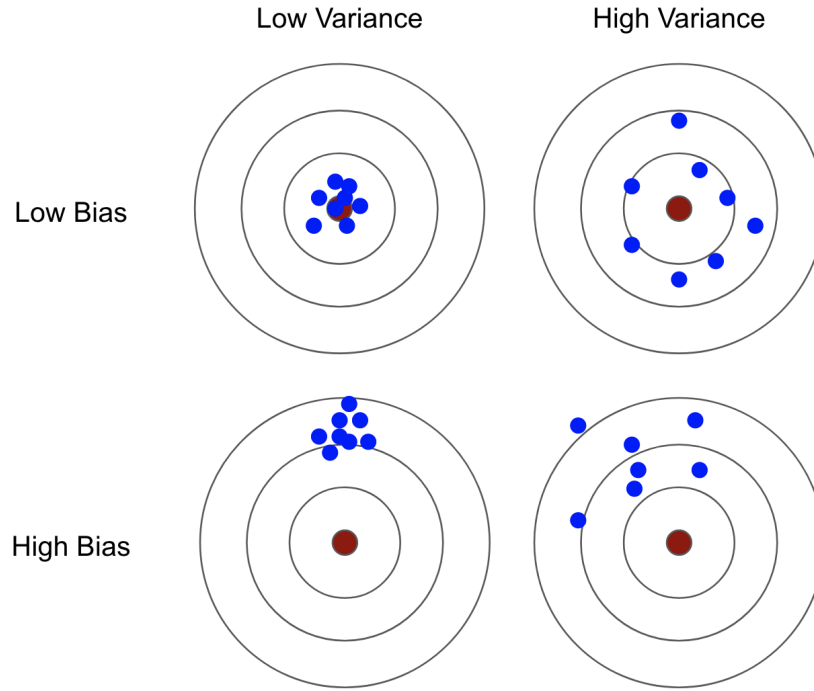


Figure 1: Visualizing the bias-variance tradeoff

Proof. Let's find the expectation of the MLE for variance. Let $\mu = \mathbb{E}[X_i]$.

$$\begin{aligned}
 \mathbb{E}[n\hat{\sigma}_{\text{MLE}}^2] &= \mathbb{E}\left[\sum_{i=1}^n (X_i - \bar{X}_n)^2\right] = \mathbb{E}\left[\sum_{i=1}^n ((X_i - \mu) - (\bar{X}_n - \mu))^2\right] \\
 &= \mathbb{E}\left[\sum_{i=1}^n ((X_i - \mu)^2 - 2(X_i - \mu)(\bar{X}_n - \mu) + (\bar{X}_n - \mu)^2)\right] \\
 &= \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] - 2\mathbb{E}\left[(\bar{X}_n - \mu) \sum_{i=1}^n (X_i - \mu)\right] + n\mathbb{E}[(\bar{X}_n - \mu)^2] \\
 &= \sum_{i=1}^n \text{Var}(X_i) - 2\mathbb{E}[(\bar{X}_n - \mu)n(\bar{X}_n - \mu)] + n\text{Var}(\bar{X}) \\
 &= n\sigma^2 - 2n\mathbb{E}[(\bar{X}_n - \mu)^2] + n\frac{\sigma^2}{n} = n\sigma^2 - 2n\frac{\sigma^2}{n} + \sigma^2 \\
 &= n\sigma^2 - 2\sigma^2 + \sigma^2 = (n-1)\sigma^2.
 \end{aligned}$$

Therefore, $\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2] = \frac{n-1}{n}\sigma^2$. The estimator is biased, as it systematically underestimates the true variance. From this proof, we can see that multiplying by $\frac{n}{n-1}$ corrects the bias, leading to the unbiased estimator S^2 . $\square$

2.2 Efficiency and the Bias-Variance Tradeoff

For unbiased estimators, we prefer the one with the smallest variance. This leads to the concept of Mean Squared Error (MSE), which combines both bias and variance.

Definition 3 (Mean Squared Error). *The MSE of an estimator is the expected squared difference between the estimator and the true parameter:*

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2.$$

*This leads to the famous **Bias-Variance Tradeoff**. Sometimes, a small amount of bias can be accepted in exchange for a large reduction in variance, leading to a lower overall MSE.*

Estimating the Bias of a Coin: This example provides a simple, concrete illustration of the bias-variance tradeoff. We will compare a standard, unbiased estimator with a slightly biased one and see how the biased estimator can sometimes be superior.

Setup: We are given a potentially biased coin and want to estimate its true probability of landing heads, denoted by p . To do this, we are allowed to flip the coin $n = 10$ times.

Suppose our experiment yields a surprising result: we observe $k = 9$ Heads and only 1 Tail. Two data analysts are asked to provide their best estimate for p .

Analyst A: The Unbiased MLE Approach: Analyst A uses the most common and intuitive estimator: the sample proportion of heads. This is the Maximum Likelihood Estimator (MLE).

- **Estimator Rule:** Let X_i be the outcome of the i -th flip ($X_i = 1$ for Heads, $X_i = 0$ for Tails). The estimator is the sample mean:

$$\hat{p}_A = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

- **Estimate for our data:** For our specific result ($k = 9, n = 10$), the estimate is:

$$\hat{p}_A = \frac{9}{10} = 0.9.$$

- **Bias Analysis:** We compute the expected value of the estimator. Since $\mathbb{E}[X_i] = p$:

$$\mathbb{E}[\hat{p}_A] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n}(np) = p.$$

The bias is $\text{Bias}(\hat{p}_A) = \mathbb{E}[\hat{p}_A] - p = p - p = 0$. This is an **unbiased** estimator.

- **Variance Analysis:**

$$\text{Var}(\hat{p}_A) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}.$$

- **Mean Squared Error (MSE):**

$$\text{MSE}(\hat{p}_A) = (\text{Bias})^2 + \text{Variance} = 0^2 + \frac{p(1-p)}{n} = \frac{p(1-p)}{10}.$$

Analyst B: The “Shrinkage” Estimator (Biased) Analyst B is skeptical of extreme results from small samples. She decides to use a “regularized” estimator that pulls the result towards the center (0.5). This estimator can be thought of as adding two “pseudo-counts”: one head and one tail.

- **Estimator Rule:**

$$\hat{p}_B = \frac{\text{Number of Heads} + 1}{n + 2} = \frac{(\sum_{i=1}^n X_i) + 1}{n + 2}.$$

- **Estimate for our data:**

$$\hat{p}_B = \frac{9 + 1}{10 + 2} = \frac{10}{12} \approx 0.833.$$

This estimate is “shrunk” towards 0.5 and seems more plausible than 0.9 for a real-world coin.

- **Bias Analysis:**

$$\begin{aligned} \mathbb{E}[\hat{p}_B] &= \mathbb{E}\left[\frac{(\sum_{i=1}^n X_i) + 1}{n + 2}\right] = \frac{\mathbb{E}[\sum_{i=1}^n X_i] + 1}{n + 2} = \frac{np + 1}{n + 2} \\ \text{Bias}(\hat{p}_B) &= \mathbb{E}[\hat{p}_B] - p = \frac{np + 1}{n + 2} - p = \frac{np + 1 - p(n + 2)}{n + 2} = \frac{1 - 2p}{n + 2}. \end{aligned}$$

This estimator is **biased** (unless $p = 0.5$). If $p < 0.5$, the bias is positive (pulling it up), and if $p > 0.5$, the bias is negative (pulling it down).

- **Variance Analysis:**

$$\text{Var}(\hat{p}_B) = \text{Var}\left(\frac{\sum_{i=1}^n X_i + 1}{n + 2}\right) = \frac{1}{(n + 2)^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{np(1-p)}{(n + 2)^2}.$$

- **Mean Squared Error (MSE):**

$$\text{MSE}(\hat{p}_B) = (\text{Bias})^2 + \text{Variance} = \left(\frac{1-2p}{n+2}\right)^2 + \frac{np(1-p)}{(n+2)^2} = \frac{(1-2p)^2 + np(1-p)}{(n+2)^2}.$$

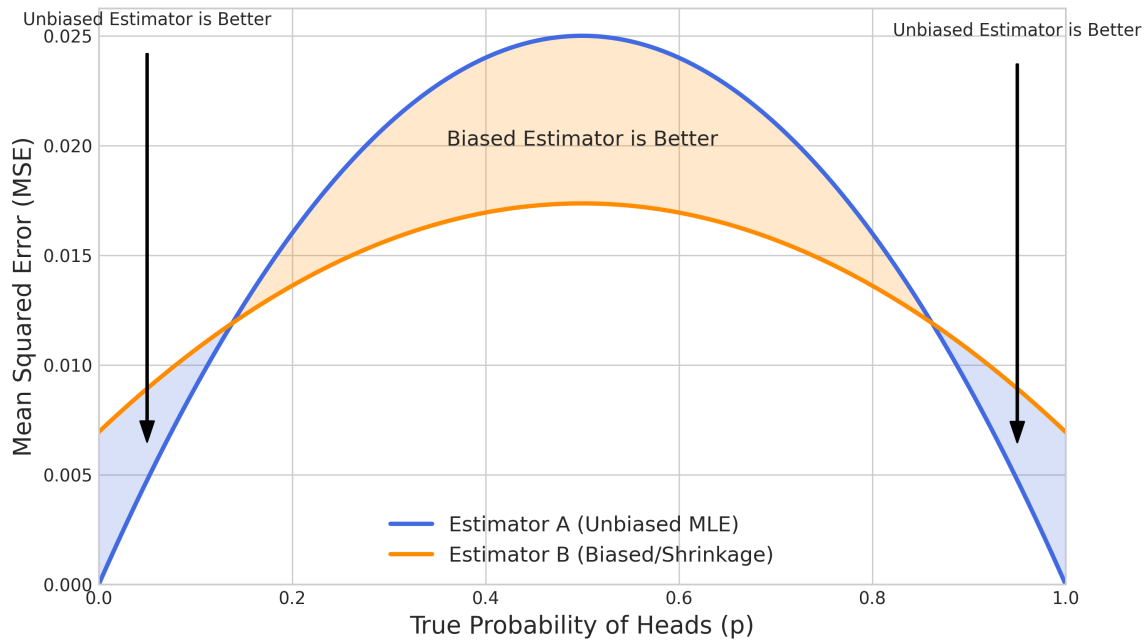
4. The Tradeoff: Comparing the MSEs

Which estimator is better? The one with the lower MSE. We compare $\text{MSE}(\hat{p}_A)$ and $\text{MSE}(\hat{p}_B)$ for $n = 10$:

$$\text{MSE}(\hat{p}_A) = \frac{p(1-p)}{10} \quad \text{vs.} \quad \text{MSE}(\hat{p}_B) = \frac{(1-2p)^2 + 10p(1-p)}{144}.$$

Instead of solving the inequality, we can analyze the components:

- **Variance Comparison:** The variance of Analyst B's estimator, $\frac{10p(1-p)}{144}$, is clearly smaller than Analyst A's, $\frac{p(1-p)}{10}$. Analyst B achieves a significant reduction in variance by increasing the denominator. Her estimator is more stable.
- **Bias Cost:** Analyst B “pays” for this lower variance by introducing bias. The bias term, $\left(\frac{1-2p}{12}\right)^2$, is small when p is near 0.5 but becomes large when p is near 0 or 1.



The plot of the two MSE functions against the true value of p reveals the tradeoff:

- **When p is in the middle range (e.g., roughly $0.2 < p < 0.8$):** The MSE of the biased estimator ($\hat{p}_B$) is lower. The large reduction in variance more than outweighs the small cost of bias. In this region, being biased is beneficial.
- **When p is extreme (near 0 or 1):** The bias of $\hat{p}_B$ becomes very large because its “prior belief” that p is near 0.5 is very wrong. The bias term dominates the MSE, making the unbiased estimator $\hat{p}_A$ superior.

The philosophy of this example can be extended in much more realistic scenarios. By introducing a small amount of bias (shrinking an estimate towards a reasonable, central value), we can often create a more stable, lower-variance estimator that has a better overall performance (lower MSE), especially for small sample sizes. This is precisely the principle behind regularization techniques like Ridge and Lasso regression in machine learning, which intentionally introduce bias to control for model variance and prevent overfitting.

2.3 Consistency

A good estimator should improve as we get more data.

Definition 4 (Consistency). An estimator $\hat{\theta}_n$ is **consistent** if it converges in probability to the true parameter as the sample size increases: $\hat{\theta}_n \xrightarrow{P} \theta$.

Proposition 2. The sample mean $\bar{X}_n$ is a consistent estimator of the population mean μ .

Proof. By definition, an estimator $\hat{\theta}_n$ is consistent if $\hat{\theta}_n \xrightarrow{p} \theta$. For the sample mean, we need to show that $\bar{X}_n \xrightarrow{p} \mu$. This requires that for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = 0.$$

This is precisely the statement of the Weak Law of Large Numbers, which holds for any i.i.d. sequence of random variables with a finite mean. Therefore, by the WLLN, the sample mean is a consistent estimator of the population mean. $\square$

The sample mean is the canonical example of a consistent estimator, with its consistency guaranteed by the Law of Large Numbers. However, this principle extends to many other important estimators. A key tool for proving this is the **Continuous Mapping Theorem**.

Theorem 3 (Continuous Mapping Theorem - for convergence in probability). *Let $\{X_n\}$ be a sequence of random variables such that $X_n \xrightarrow{p} c$ for some constant c . If g is a function that is continuous at the point c , then:*

$$g(X_n) \xrightarrow{p} g(c).$$

This theorem allows us to establish the consistency of estimators that are functions of other consistent estimators.

The Unbiased Sample Variance: Let $X_1, \dots, X_n$ be an i.i.d. sample from a distribution with mean μ and variance σ^2 . The unbiased sample variance is a consistent estimator of the true variance σ^2 :

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \xrightarrow{p} \sigma^2.$$

Proof Sketch. First, we rewrite the estimator to make it more convenient:

$$S_n^2 = \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2 \right).$$

We analyze each component's convergence:

Convergence of the first term: The variables X_i^2 are also i.i.d. By the Law of Large Numbers applied to this new sequence, their sample mean converges in probability to their expected value:

$$\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{p} \mathbb{E}[X^2].$$

Convergence of the second term: We already know from the WLLN that the sample mean is consistent:

$$\bar{X}_n \xrightarrow{p} \mathbb{E}[X] = \mu.$$

By the Continuous Mapping Theorem with the function $g(x) = x^2$, we have:

$$(\bar{X}_n)^2 \xrightarrow{p} \mu^2 = (\mathbb{E}[X])^2.$$

Combining the terms: We can use another property of convergence in probability (Slutsky's Theorem, though the Continuous Mapping Theorem for a function of two variables also works) to combine these results. The expression in the parenthesis converges:

$$\left(\frac{1}{n} \sum_{i=1}^n X_i^2 - (\bar{X}_n)^2 \right) \xrightarrow{p} \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \sigma^2.$$

The pre-factor: The term $\frac{n}{n-1}$ is a deterministic sequence that converges to 1 as $n \rightarrow \infty$.

$$\lim_{n \rightarrow \infty} \frac{n}{n-1} = 1.$$

Multiplying the two converging parts together, we conclude that the entire expression converges to $1 \cdot \sigma^2 = \sigma^2$. Thus, the unbiased sample variance is a consistent estimator of the true variance. $\square$

Method of Moments Estimator for the Exponential Distribution: In the previous section, we found the Method of Moments estimator for the rate parameter λ of an Exponential distribution to be:

$$\hat{\lambda}_n = \frac{1}{\bar{X}_n}.$$

This estimator is consistent.

Proof. This is a very direct application of the WLLN and the Continuous Mapping Theorem. By the Law of Large Numbers, the sample mean $\bar{X}_n$ is a consistent estimator of the true mean $\mathbb{E}[X]$:

$$\bar{X}_n \xrightarrow{p} \mathbb{E}[X].$$

For an $\text{Exp}(\lambda)$ distribution, the true mean is $\mathbb{E}[X] = 1/\lambda$. So, $\bar{X}_n \xrightarrow{p} 1/\lambda$. Our estimator is a function of the sample mean, $\hat{\lambda}_n = 1/\bar{X}_n = g(\bar{X}_n)$, where the function is $g(x) = 1/x$. Since the true mean $1/\lambda$ is non-zero, the function $g(x) = 1/x$ is continuous at this point. By the Continuous Mapping Theorem, we can apply the function g to the convergence result:

$$\begin{aligned} g(\bar{X}_n) &\xrightarrow{p} g(1/\lambda) \\ \frac{1}{\bar{X}_n} &\xrightarrow{p} \frac{1}{1/\lambda} = \lambda. \end{aligned}$$

Therefore, $\hat{\lambda}_n$ is a consistent estimator of λ . This logic applies to most Method of Moments estimators. $\square$

Maximum Likelihood Estimators (General Property): While the proofs are more advanced and require certain “regularity conditions” on the underlying probability distribution, one of the most important theoretical results in statistics is that **Maximum Likelihood Estimators are generally consistent**, i.e. if $\hat{\theta}_{\text{MLE}}$ is the MLE for a parameter θ , then:

$$\hat{\theta}_{\text{MLE}} \xrightarrow{p} \theta.$$

This powerful property is a key reason why MLE is the most widely used method for parameter estimation. It provides a strong theoretical guarantee that, given enough data, the MLE will get arbitrarily close to the true parameter value.

3 The Cramér-Rao Lower Bound

3.1 Motivation: A Benchmark for Estimators

We have seen that for unbiased estimators, we prefer the one with the smallest variance. But this raises a fundamental question: what is the *best possible* variance we can ever hope to achieve? Is there a theoretical limit to how well we can estimate a parameter from data?

The Cramér-Rao Lower Bound (CRLB) provides this limit. It establishes a lower bound on the variance of *any* unbiased estimator, giving us a benchmark against which we can measure the performance of our estimators. An estimator that achieves this bound is, in a very strong sense, the best possible.

3.2 Fisher Information

To establish this bound, we first need to quantify how much “information” a single observation X provides about an unknown parameter θ . This concept is called **Fisher Information**.

Intuitively, if the log-likelihood function $\ell(\theta) = \log f(x; \theta)$ is very sharply peaked around its maximum, then the data provides a lot of information; a small change in θ leads to a large drop in likelihood. If the log-likelihood is flat, the data is less informative. The curvature of the log-likelihood is therefore related to information.

Definition 5 (Fisher Information). Let X be a random variable with PDF (or PMF) $f(x; \theta)$. The **Fisher Information** that X carries about θ , denoted $I(\theta)$, is defined as the variance of the score function (the derivative of the log-likelihood):

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right].$$

Under suitable regularity conditions (allowing us to exchange differentiation and integration), this can also be expressed as the negative expected value of the second derivative:

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right].$$

For an i.i.d. sample of size n , the Fisher Information of the entire sample, $I_n(\theta)$, is simply n times the information of a single observation: $I_n(\theta) = n \cdot I(\theta)$.

3.3 The Cramér-Rao Lower Bound (CRLB)

We can now state the main theorem.

Theorem 4 (The Cramér-Rao Lower Bound). *Let $X_1, \dots, X_n$ be an i.i.d. sample from a distribution $f(x; \theta)$. Let $\hat{\theta}$ be any **unbiased** estimator of θ ($\mathbb{E}[\hat{\theta}] = \theta$). Under regularity conditions, the variance of $\hat{\theta}$ is bounded below by the reciprocal of the Fisher Information:*

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I_n(\theta)} = \frac{1}{n \cdot I(\theta)}.$$

Definition 6 (Efficient Estimator). *An unbiased estimator $\hat{\theta}$ is called an **efficient estimator** if its variance achieves the Cramér-Rao Lower Bound. That is, if:*

$$\text{Var}(\hat{\theta}) = \frac{1}{n \cdot I(\theta)}.$$

An efficient estimator is, in this sense, a “best unbiased estimator”.

Efficiency of the Sample Mean for a Normal Distribution: Let's prove that the sample mean, $\bar{X}_n$, is the most efficient unbiased estimator for the mean μ of a Normal distribution $N(\mu, \sigma^2)$, assuming σ^2 is known.

1. Write the log-likelihood for a single observation X :

$$\log f(x; \mu) = \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right) \right) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x - \mu)^2}{2\sigma^2}.$$

2. Compute the derivatives with respect to μ :

• First derivative (score):

$$\frac{\partial}{\partial \mu} \log f(x; \mu) = -\frac{2(x - \mu)(-1)}{2\sigma^2} = \frac{x - \mu}{\sigma^2}$$

• Second derivative (Hessian):

$$\frac{\partial^2}{\partial \mu^2} \log f(x; \mu) = -\frac{1}{\sigma^2}.$$

3. Compute the Fisher Information $I(\mu)$: Using the second-derivative form is easier here, as the result is a constant and does not depend on x .

$$I(\mu) = -\mathbb{E} \left[\frac{\partial^2}{\partial \mu^2} \log f(X; \mu) \right] = -\mathbb{E} \left[-\frac{1}{\sigma^2} \right] = \frac{1}{\sigma^2}.$$

4. Determine the Cramér-Rao Lower Bound: For a sample of size n , the Fisher Information is $I_n(\mu) = n \cdot I(\mu) = \frac{n}{\sigma^2}$. The CRLB for any unbiased estimator $\hat{\mu}$ is therefore:

$$\text{Var}(\hat{\mu}) \geq \frac{1}{I_n(\mu)} = \frac{1}{n/\sigma^2} = \frac{\sigma^2}{n}.$$

5. Compare with the Sample Mean: We already know that the sample mean $\bar{X}_n$ is an unbiased estimator for μ . We also know that its variance is:

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Since the variance of our estimator exactly equals the Cramér-Rao Lower Bound, we can conclude that $\bar{X}_n$ is an **efficient estimator**. It is, provably, the best possible unbiased estimator for the mean of a Normal distribution.